中国海洋大学海德学院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Week 6

Outline:

i. Polynomial Regression

ii. Estimation of ANOVA

Multiple Regression model/GLM

Multiple regression/GLM is "linear in the regressors (or independent variables)"

The predicted score is a linear combination of the regressors (X's) in the model

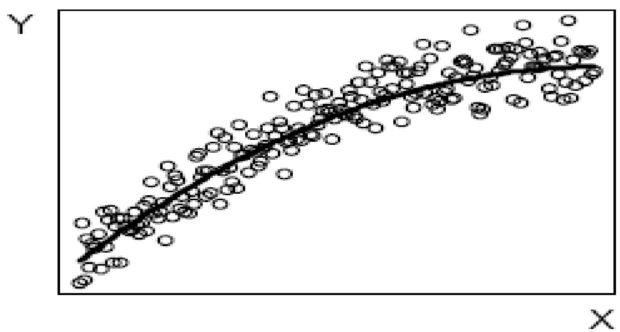
Each regressor is multiplied by its coefficient and added together (+ intercept/constant)

$Y = b_0 + b_1X_1 + b_2X_2 + \ldots\ldots$

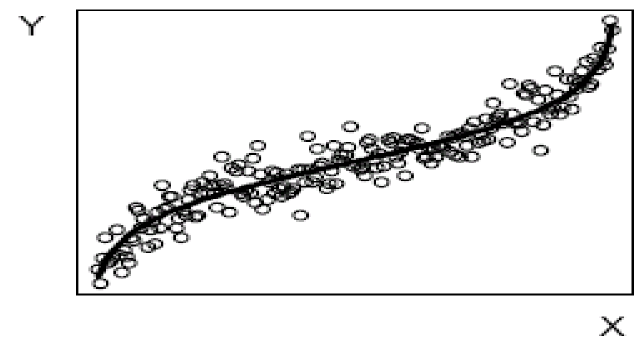
How to Handle Non-Linear Effects

Power transformations can make simple monotone relationships more linear (Fig a). Polynomial regression (or other transformations, e.g. logit) is often needed for more complex relationships (Figs, b & c)

(a)



(b)



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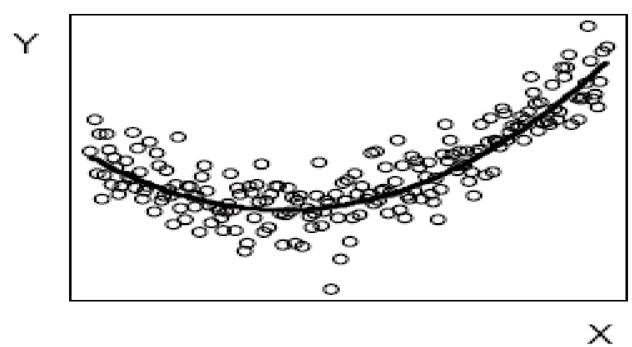


How to Handle Non-Linear Effects

- Simple monotonic relationships: Power transformations.
- Polynomial Regression- Quantitative Variables
- Polynomial Regression- Categorical Variables
- Generalized Linear Models (e.g., logistic regression)

Linear vs. Logistic Regression

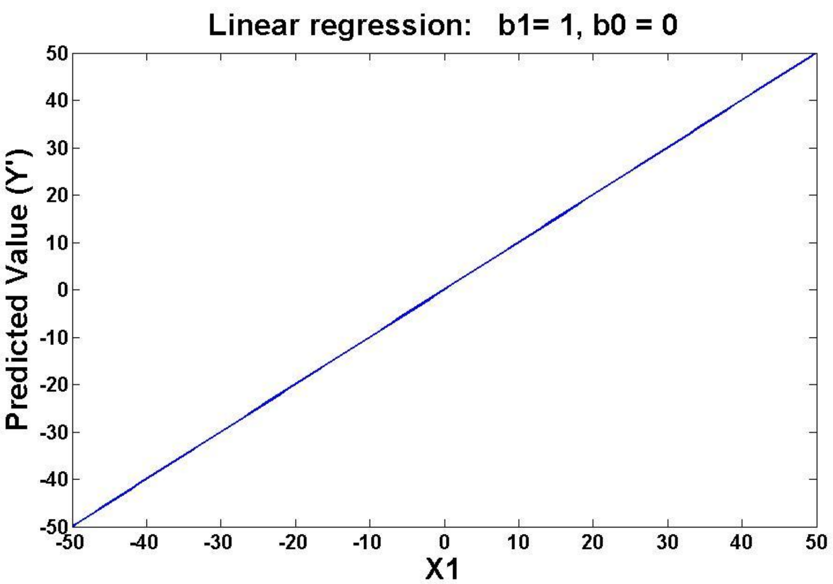
Linear Regression

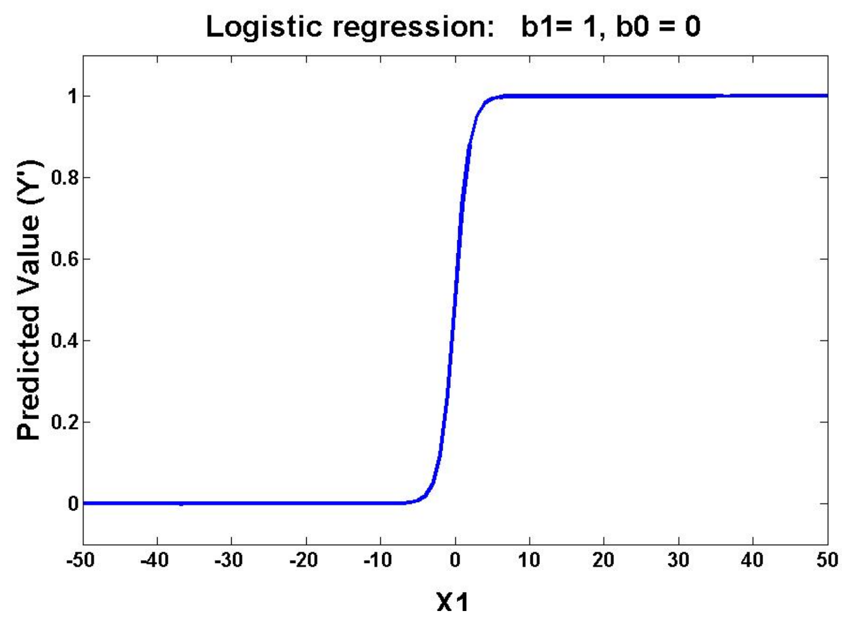
$$Y = b_0 + b_1X_1$$

Logistic Regression

$$Y = \frac{e^{b0 + b1X1}}{(1 + e^{b0 + b1X1})}$$

Predicted values graphed for X1 = -50 to 50





Polynomial Regression

$Y = A + BX + CX^2 + DX^3 + \ldots.. + QX^{N-1}$

It is important to distinguish between regressors in the model vs. variables of interest.

In this example there is only one variable of interest. The powers of X act as a structural set of regressors to allow for non-linear relationships between this and variable of interest Y.

However, the model is still linear in the parameters. I.E. linear combination of the regressors multiplied by their parameter estimates.

Polynomial Regression Order

$Y = A + BX + CX^2 + DX^3 + \ldots.. + QX^{N-1}$

If you include (N-1) regressors based on X, you will perfectly fit the data.

The order of the equation is the highest power: (N-1) in this example.

$X^{(N-1)}$ is the highest order predictor. All other regressors are lower order.

The highest order regressor determines the overall shape of the relationship within the range of $(-\infty$ to $\infty)$

Polynomial Regression Shape

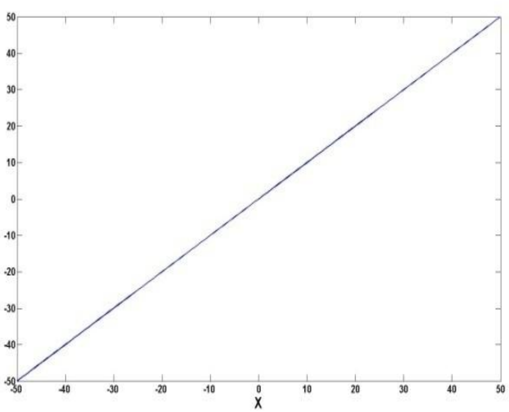
$Y = A + BX + CX^2 + DX^3 + \ldots\ldots QX^{N-1}$

The highest order regressor determines the overall shape of the relationship within the range of –inf to inf

Linear

$Y = A + BX$

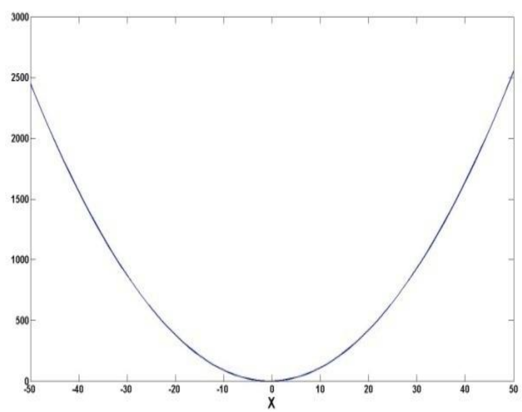
Zero bends



Quadratic

$Y = A + BX + CX^2$

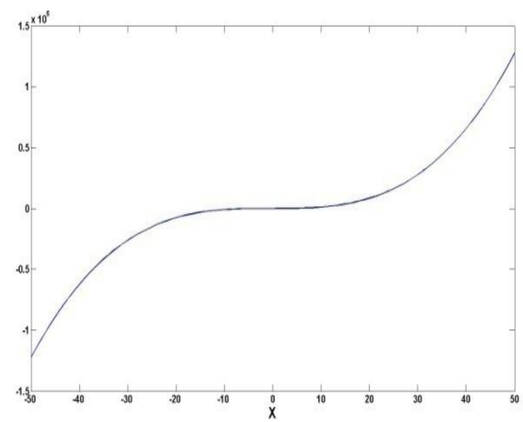
One bend



Cubic

$Y = A + BX + CX^2 + DX^3$

Two bends



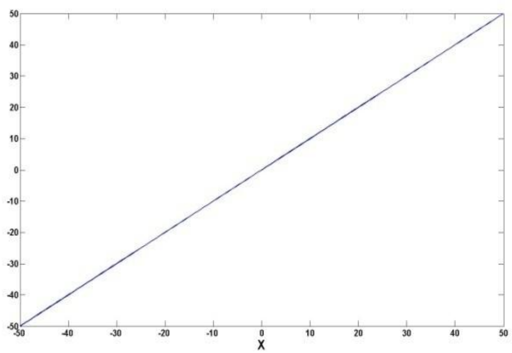
There is one less bend than the highest order in the polynomial model.

Shape and Coefficient Sign

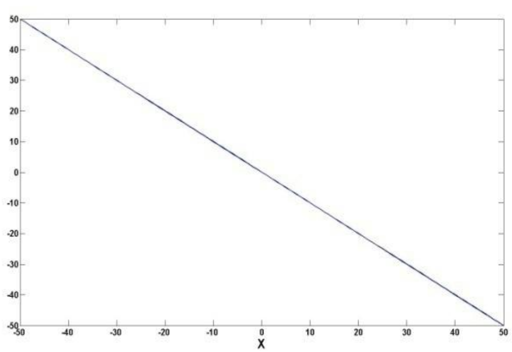
The sign of the coefficient for the highest order regressor determines the direction of the curve.

Linear

$Y = 0 + 1X$

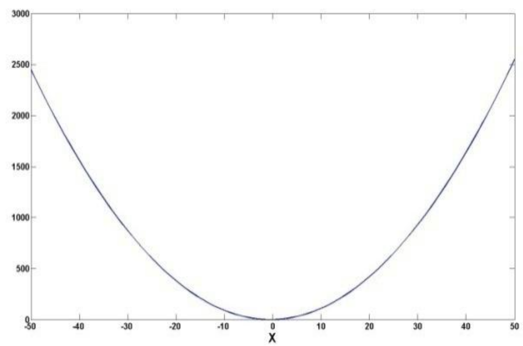


$Y = 0 + -1X$

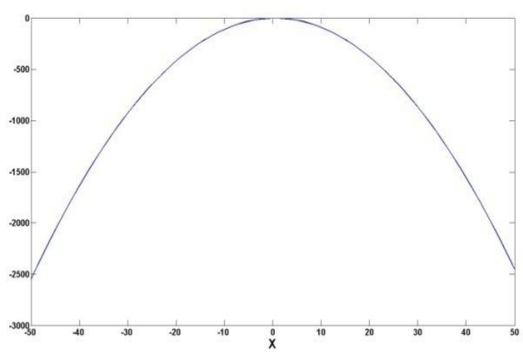


Quadratic

$Y = 0 + 1X + 1X^2$

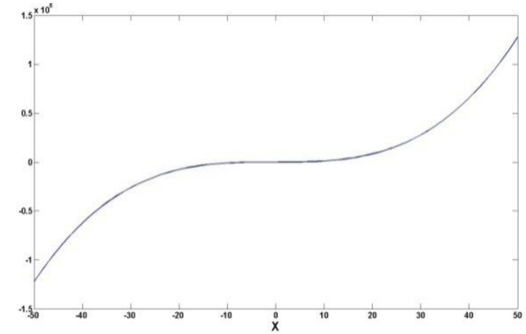


$Y = 0 + 1X + -1X^2$

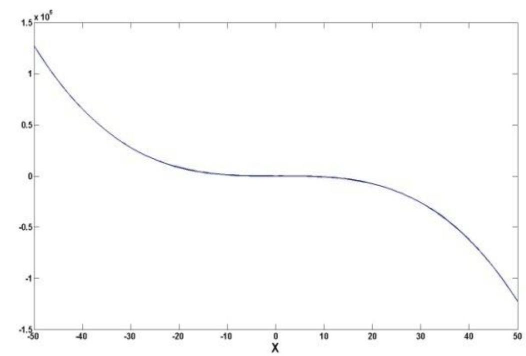


Cubic

$Y = 0 + 1X + 1X^2 + 1X^3$

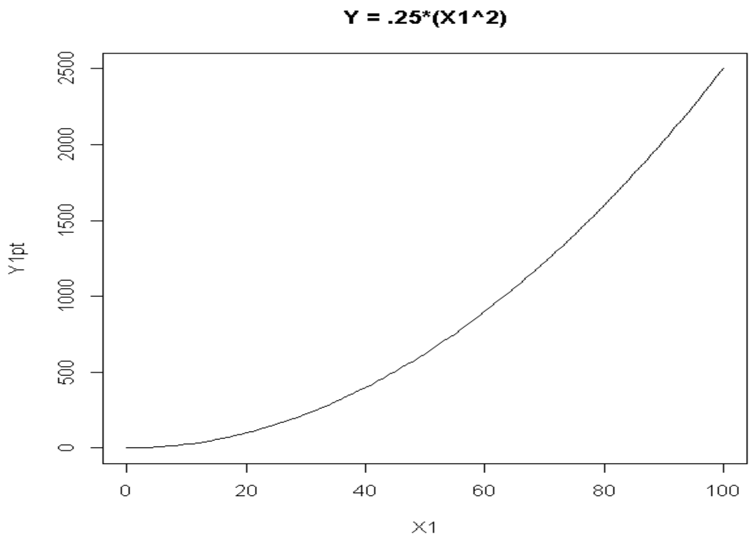


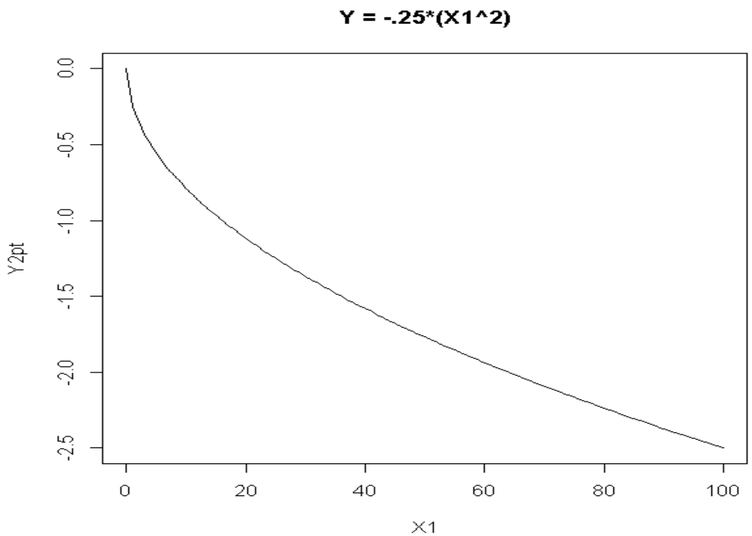
$Y = 0 + 1X + 1X^2 + -1X^3$



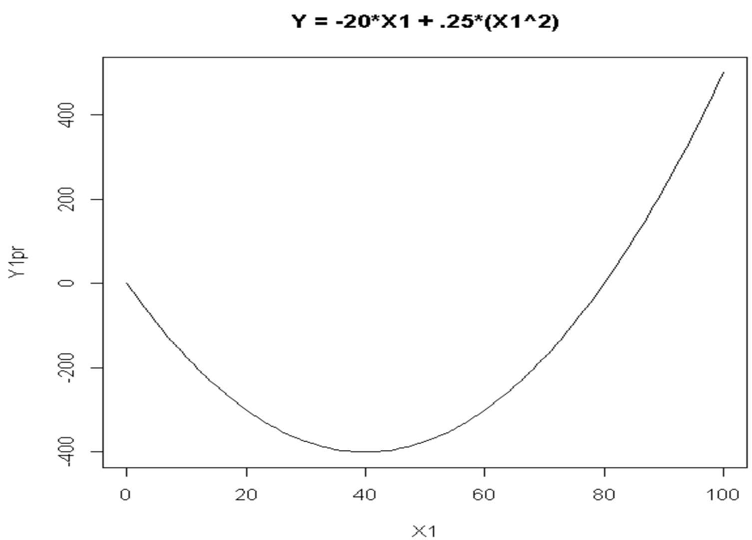
Change of Sign in Power Transformation and Polynomial

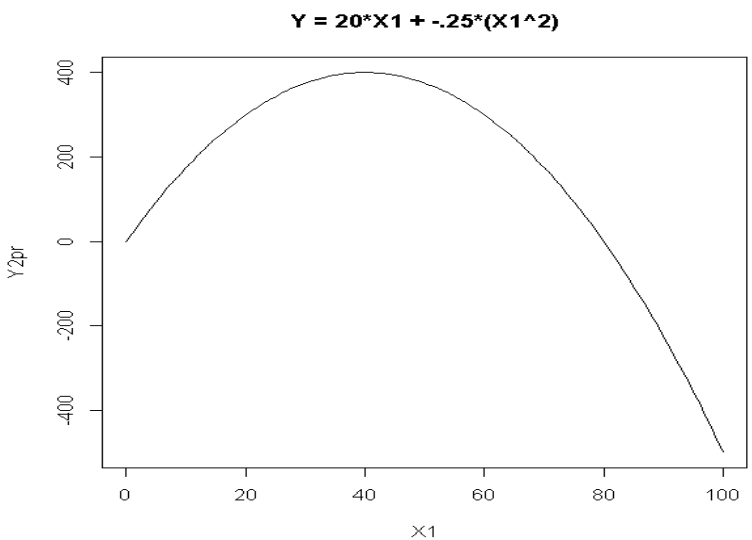
Power Transformations of X





Polynomial Regression





Example 1: Predicting Interest in Electives

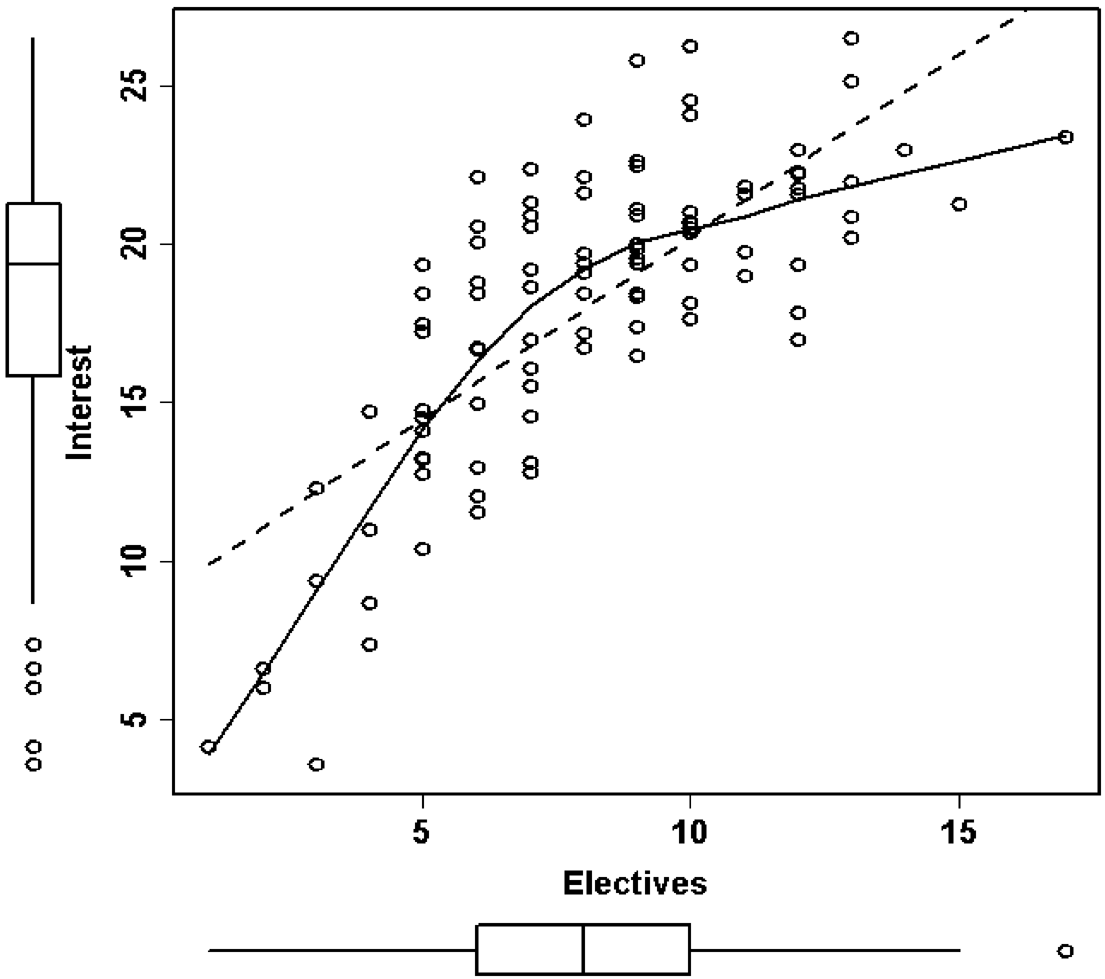
How does number of electives (X) taken in an area predict interest (Y) in further electives?

```
varDescribe(dE)
          var   n  mean   sd median trimmed  mad  min   max range  skew
Electives   1 100  8.17 3.09   8.00    8.15 2.97 1.00 17.00 16.00  0.14
Interest    2 100 18.10 4.75  19.33   18.58 3.62 3.56 26.49 22.92 -0.96

         kurtosis   se
Electives    -0.30 0.31
Interest      0.74 0.47

cor(dE)
          Electives  Interest
Electives 1.0000000 0.7488801
Interest  0.7488801 1.0000000
```

Example 1: Predicting Interest in Electives

Scatter plot

```
scatterplot(dE$Electives, dE$Interest,cex=1.5, lwd=2,xlab = 'Electives',
               ylab='Interest', col='black')
```

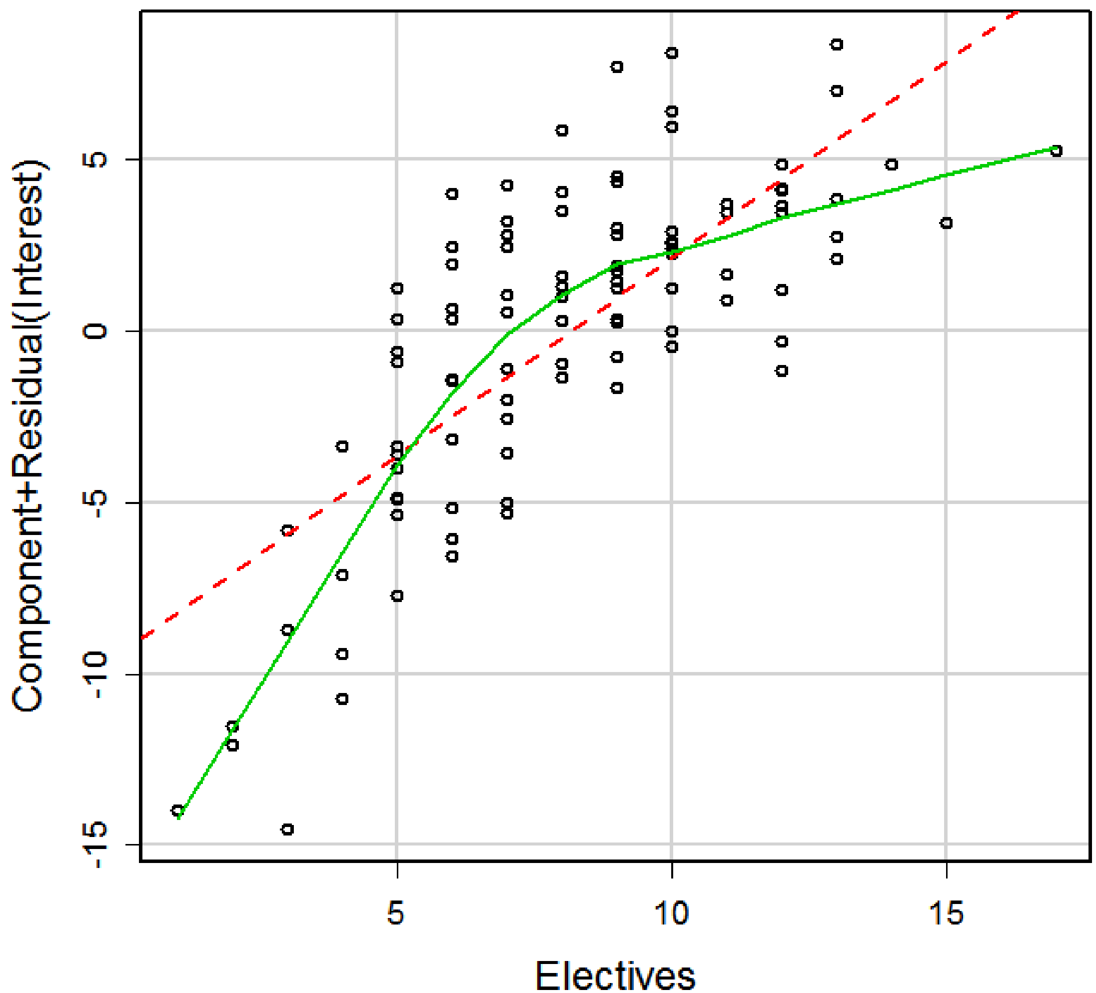


Example 1: Predicting Interest in Electives

```
mLinear = lm(Interest ~ Electives, data=dE)
summary(mLinear)

Coefficients
              Estimate   SE    t-statistic Pr(>|t|)
(Intercept)    8.7177 0.8968       9.721 4.87e-16 ***
Electives      1.1490 0.1027      11.187  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sum of squared errors (SSE): 980.2, Error df: 98
R-squared:  0.5608
```

Example 1: Predicting Interest in Electives

```
modelAssumptions(mLinear,Type='linear', one.page=FALSE)
```



Example 1: Predicting Interest in Electives

```
mLinear = lm(Interest ~ Electives, data=dE)
summary(mLinear)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   8.7177     0.8968   9.721 4.87e-16
Electives     1.1490     0.1027  11.187  < 2e-16

Residual standard error: 3.163 on 98 degrees of freedom
Multiple R-squared: 0.5608,     Adjusted R-squared: 0.5563
F-statistic: 125.1 on 1 and 98 DF,  p-value: < 2.2e-16
```

What are your concerns about interpreting this model?

The linearity assumption regarding the electives effect is pretty clearly violated. When this assumption is violated, the parameter estimates are biased. In this case, you can expect that the model is underestimating the true magnitude of the electives effect. This is a bad model or not best fitted model.

How to estimate polynomial (quadratic) model

```
dE$Electives2 = dE$Electives * dE$Electives
mQuad= lm(Interest ~ Electives + Electives2, data=dE)
summary(mQuad)
Coefficients
            Estimate    SE    t-statistic          Pr(>|t|)
(Intercept)   1.0031  1.5503       0.647              0.519
Electives     3.2743  0.3800       8.617 0.00000000000129 ***
Electives2   -0.1266  0.0220      -5.754 0.00000101764858 ***
---
Sum of squared errors (SSE): 730.8, Error df: 97
R-squared:  0.6726
```

Or

```
mQuad= lm(Interest ~ Electives + I(Electives^2), data=dE)
summary(mQuad)
Coefficients
               Estimate      SE t-statistic          Pr(>|t|)
(Intercept)      1.0031  1.5503       0.647             0.519
Electives        3.2743  0.3800       8.617 0.00000000000129 ***
I(Electives^2)  -0.1266  0.0220      -5.754 0.00000101764858 ***
---
Sum of squared errors (SSE): 730.8, Error df: 97
R-squared:  0.6726
```

Compare linear and quadratic models

Quadratic model

```
mQuad = lm(Interest ~ Electives + I(Electives^2), data=dE)
summary(mQuad)
Coefficients
               Estimate      SE   t-statistic           Pr(>|t|)
(Intercept)      1.0031  1.5503         0.647              0.519
Electives        3.2743  0.3800         8.617 0.000000000000129 ***
I(Electives^2)  -0.1266  0.0220        -5.754 0.000000101764858 ***
---

Sum of squared errors (SSE): 730.8, Error df: 97
R-squared:  0.6726
```

Linear model

```
mLinear = lm(Interest ~ Electives, data=dE)
summary(mLinear)
Coefficients
            Estimate     SE   t-statistic Pr(>|t|)
(Intercept)   8.7177 0.8968         9.721 4.87e-16 ***
Electives     1.1490 0.1027        11.187  < 2e-16 ***
---

Sum of squared errors (SSE): 980.2, Error df: 98
R-squared:  0.5608
```

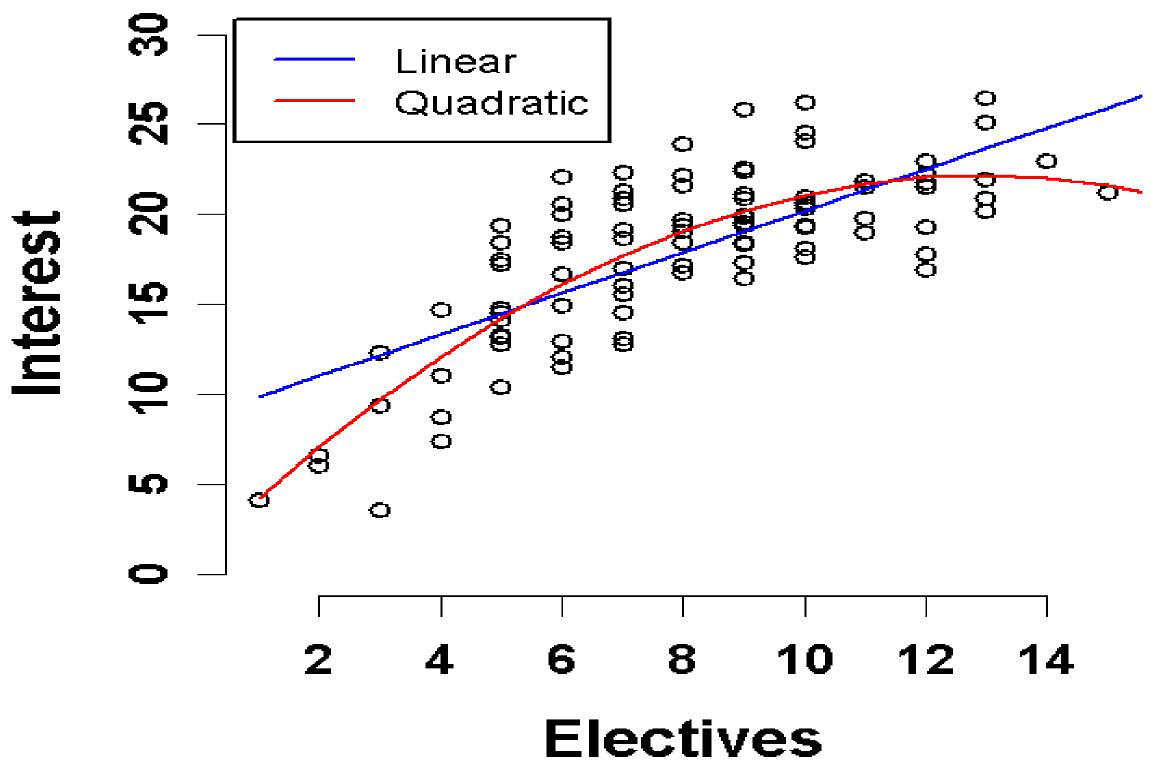
Example 1: Predicting Interest in Electives

Linear model

Interest = <u>8.7</u> + <u>1.4</u>*Electives

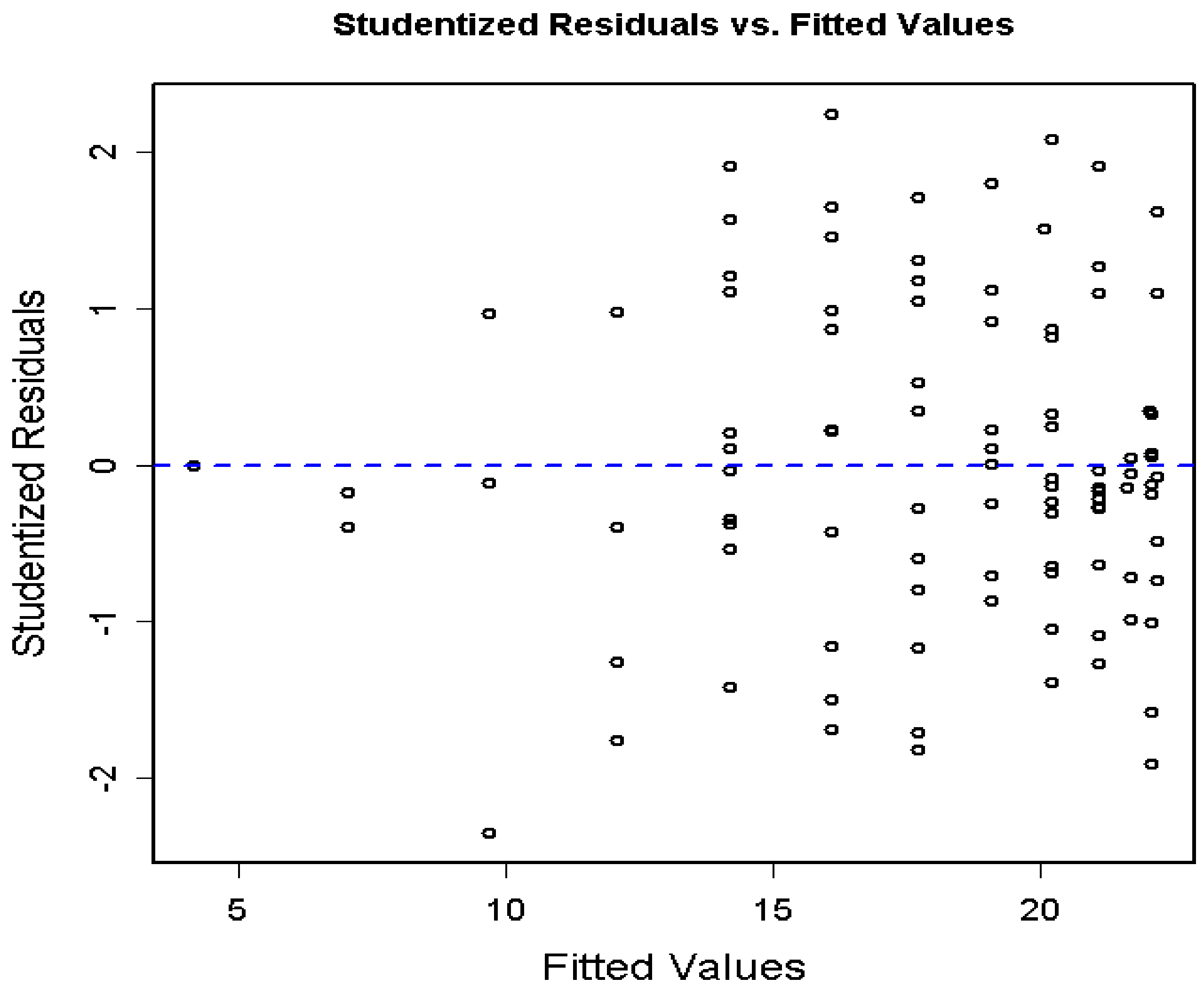
Quadratic model

Interest = <u>1.0</u> + <u>3.3</u>*Electives + <u>-0.1</u>*Electives^2



Example 1: Predicting Interest in Electives

The Standardized Residuals plot



How to test overall effect of variable

```
mQuad = lm(Interest ~ Electives + I(Electives^2), data=dE)
summary(mQuad)
Coefficients
                    Estimate     SE     t-statistic            Pr(>|t|)
(Intercept)           1.0031  1.5503        0.647                0.519
Electives             3.2743  0.3800        8.617 0.000000000000129 ***
I(Electives^2)       -0.1266  0.0220       -5.754 0.000000101764858 ***
---

Sum of squared errors (SSE): 730.8, Error df: 97
R-squared:  0.6726
```

How do we get a test of the overall Electives effect in polynomial (i.e., quadratic) model?

The test of the set of regressors that code for Electives, In this case, Electives and Electives^2 provides the overall test. Can use R^2 here b/c these are the only two regressors in the model.

Is quadratic model better

```
mQuad = lm(Interest ~ Electives + I(Electives^2), data=dE)
summary(mQuad)
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.0031     1.5503   0.647    0.519
Electives        3.2743     0.3800   8.617 1.29e-13
I(Electives^2)  -0.1266     0.0220  -5.754 1.02e-07

Residual standard error: 2.745 on 97 degrees of freedom
Multiple R-squared: 0.6726,     Adjusted R-squared: 0.6658
F-statistic: 99.62 on 2 and 97 DF,  p-value: < 2.2e-16
```

How can you test if the quadratic model is necessary?

Test if the quadratic model fits better than the linear model.

You can compare the augmented model with electives and electives^2 to the compact model with only electives. This is of course, equivalent to testing if the coefficient for electives^2 is 0.

Is higher order model necessary

How can you test if more complex model (e.g., cubic) is needed?

Fit the cubic and compare this augmented cubic model to quadratic model or test if the coefficient for cubic term is 0.

```
mCubic = lm(Interest ~ Electives + I(Electives^2) + I(Electives^3),
            data=dE)
summary(mCubic)

Coefficients
                Estimate         SE t-statistic   Pr(>|t|)
(Intercept)    -2.581884  2.485665      -1.039    0.30155
Electives       4.937010  0.982495       5.025 0.00000233 ***
I(Electives^2) -0.342747  0.120023      -2.856    0.00526 **
I(Electives^3)  0.008285  0.004524       1.831    0.07015 .
---

Sum of squared errors (SSE): 706.1, Error df: 96
R-squared:  0.6836
```

Example 1: Interpreting the Coefficients

Linear model

Interest = <u>8.7</u> + <u>1.4</u>*Electives

Quadratic model

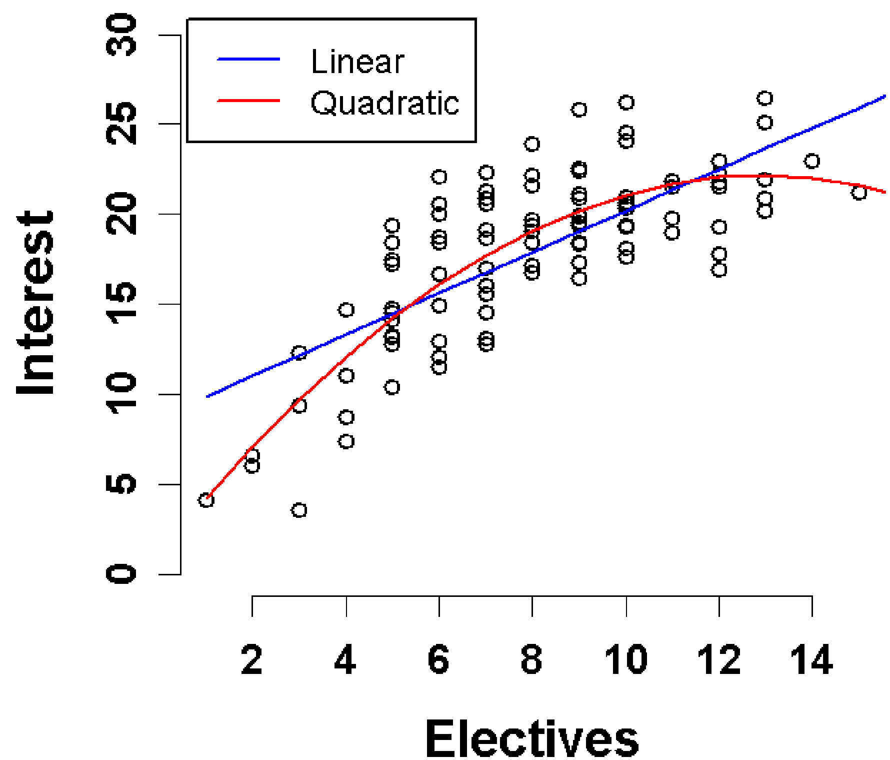
Interest = <u>1.0</u> + <u>3.3</u>*Electives + <u>-0.1</u>*Electives^2

<u>In polynomial model</u>

b_0 interpretation is unchanged but its value will likely change. Predicted value when Electives = 0

b_1 is the linear effect of Electives at Electives = 0

Higher order terms inform us how the linear effect changes across distribution of X



Details

Regressors variables in polynomial regression will be correlated unless X is centered and perfectly symmetric (or orthogonal coefficients and equal N with categorical polynomial regression)

Therefore, the current interpretation of higher order regressors applies only if all lower order predictors are partial led (i.e., included in the model).

Lower order regressors become "simple effects". E.g.- Linear effect of X at X=0.

As with interactions, you can get simple (linear) effect of the variable at any point along its distribution by centering itself on that raw score and fitting a new model.

Simple Effects (Slopes)

You can also derive the formula to describe how the simple slope changes across X using calculus (partial derivative).

We want:

$$\frac{\Delta Y}{\Delta X_i}$$

at a specific value of X_i

Use these three simple rules:

1. The partial derivative of a sum, with respect to X_i, equals the sum of the partial derivatives of the components of that sum.
2. The partial derivative of aX_i^M, with respect to X_i is aMX_i^{M-1}, where *a* can be either a constant or another variable (or some combination of both
3. The partial derivative of a component of a sum, with respect to X_i, where that component does not contain X_i, is zero

Simple Effects (Slopes)

1. The partial derivative of a sum, with respect to X_i, equals the sum of the partial derivatives of the components of that sum.

2. The partial derivative of aX_i^M, with respect to X_i is aMX_i^{M-1}, where *a* can be either a constant or another variable (or some combination of both

3. The partial derivative of a component of a sum, with respect to X_i, where that component does not contain X_i, is zero

Interest = 1.00 + 3.27*Electives + -0.13*Electives^2

What is the formula that describes the magnitude of the Elective effect on interest (ΔInterest/ ΔElectives) across the range of electives?

ΔInterest/ ΔElectives = 3.27 – 0.26 * Electives

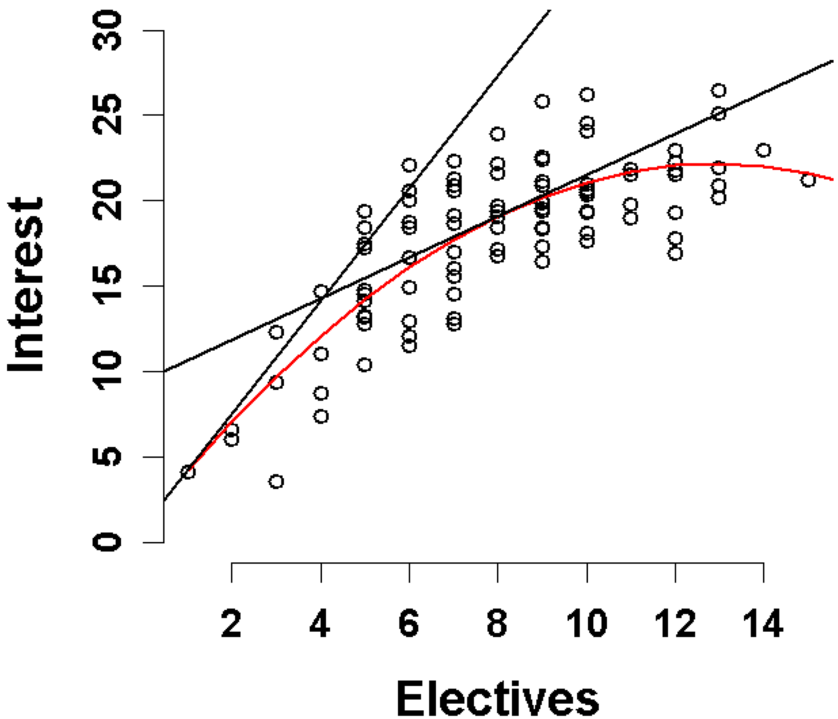
Simple Effects (Slopes)

Interest = 1.0 + 3.27*Electives + -0.13*Electives^2

ΔInterest/ ΔElectives = 3.27 – 0.26 * Electives

For Electives = 0: ΔInterest/ ΔElectives = 3.27 – 0.26 * (0) = 3.3

For Electives = 8.17 : ΔInterest/ ΔElectives = 3.27 – 0.26 * (8.17) = 1.2



Conditional effect of Xi

1. The partial derivative of a sum, with respect to X_i, equals the sum of the partial derivatives of the components of that sum.

2. The partial derivative of a X_i^M, with respect to X_i is aMX_i^{M-1}, where *a* can be either a constant or another variable (or some combination of both

3. The partial derivative of a component of a sum, with respect to X_i, where that component does not contain X_i, is zero

$Y = 2 + 3X_1$

What is the formula that describes the magnitude of the X_1 effect on Y ($\Delta Y / \Delta X_1$) and why does this make sense?

$\Delta Y / \Delta X1 = 3$ This makes sense because this is a simple linear model and the effect of X1 is the same across the whole range of X1 values.

Conditional effect of Xi

1. The partial derivative of a sum, with respect to Xi, equals the sum of the partial derivatives of the components of that sum.

2. The partial derivative of aXiM, with respect to Xi is aMXiM-1, where a can be either a constant or another variable (or some combination of both

3. The partial derivative of a component of a sum, with respect to Xi, where that component does not contain Xi, is zero
4. Y = 7 + 5X1 + 4X2
5. What is the formula that describes the magnitude of the X1 effect on Y (ΔY/ ΔX1) across the range of X1 scores and why does this make sense?

ΔY/ ΔX1 = 5 This makes sense because this is an additive model and the effect of X1 is the same across the whole range of X1 and X2 values. Similarly, ΔY/ ΔX2 = 4

Simple Effects (Slopes): Interactions

1. The partial derivative of a sum, with respect to X_i, equals the sum of the partial derivatives of the components of that sum.

2. The partial derivative of aX_i^M, with respect to X_i is aMX_i^{M-1}, where *a* can be either a constant or another variable (or some combination of both

3. The partial derivative of a component of a sum, with respect to X_i, where that component does not contain X_i, is zero

BC Intent= -1 + 6*Attitudes + 1*Peer Pressure + -1*ATTxPP

What is the formula the describes the magnitude of the Attitudes effect on BC Intent (ΔBC Intent/ ΔAttitude)

ΔBC Intent/ΔAttitudes = 6 - 1*Peer Pressure

Simple Effects (Slopes): Interactions!!!!

1. The partial derivative of a sum, with respect to X_i, equals the sum of the partial derivatives of the components of that sum.

2. The partial derivative of aX_i^M, with respect to X_i is aMX_i^{M-1}, where *a* can be either a constant or another variable (or some combination of both

3. The partial derivative of a component of a sum, with respect to X_i, where that component does not contain X_i, is zero

BC Intent= -1 + 6*Attitudes + 1*Peer Pressure + -1*ATTxPP

What is the formula the describes the magnitude of the Peer pressure effect on BC Intent (ΔBC Intent/ ΔPeer pressure)

ΔBC Intent/ΔPeer Pressure = 1 - 1*Attitudes

Sample Results Section

We regressed Interest on regressors that modeled the linear and quadratic effects of Number of Electives. Number of Electives was mean-centered in the primary analyses. We report raw regression coefficients (Bs) and partial eta^2 (η_p^2), as appropriate, to quantify effect sizes.

[This is paragraph optional in some cases]
The overall effect of Number of Electives was significant, $F(2,97) = 99.62$, $p < .0001$, with Number of Electives accounting for 67.3% of the total variance in Interest.

The linear effect of Electives was significant, B= 1.2, η_p^2= .651, t(97)= 13,45, p < .0001, indicating that taking an additional elective was associated with at 1.2 point increase in interest for participants who had already taken an average number of electives. However, the quadratic effect of Electives was also significant, B=-0.1, t(97)= 5.75, p < .0001, indicating that the magnitude of the Electives effect decreased by .02 for every additional Elective taken.

[This is paragraph optional in some cases]
Formal testing of simple effects of Electives across meaningful values for Electives indicated that the magnitude of the Electives effects was significant for participants who had taken no previous electives, B= 3.3, η_p^2= .434, t(97)= 8.62, p< .0001. The effect of Electives was also significant for participants who had taken a low number of electives (i.e., mean - 1 SD; 5.1 electives), B= 2.0, η_p^2= .582, t(97)= 11.63, p< .0001 and a high number of electives (i.e., mean + 1 SD; 11.3 electives), B= 0.4, η_p^2= .072, t(97)= 2.73, p= .0074.

Example 2: Multiple Regression with Non-Linear Effects

The non-linear effects of individual variables can be evaluated in models that contain other variables as well.

For example, lets examine the effect of Age and Weekly training miles on cross country skiers 5K race time. We might expect that there are diminishing returns with increasing weekly mileage and at some point, more miles may even hurt.

Describe the expected relationship between Miles and 5K times?

We would expect a non-linear relationship between Miles and 5K. The relationship should generally be negative with an increase in miles leading to a decrease in 5K times. However, the magnitude of the decrease in 5K Time per Mile increase will not be constant. The magnitude of this effect will decrease across the distribution of Miles.

Why would we include Age as an additional predictor in this model (4 reasons)?

1. To simultaneously study Age effects in the same sample.
2. To use Age as a covariate to increase power to test Miles effect.
3. To use Age as a covariate to examine "unique" effect of Miles controlling for Age.
4. To test for an interaction between Age and Miles.

Multiple Regression with Non-Linear Effects

Describe how to test the non-linear (polynomial) additive effects of Miles on 5K Times, controlling for Age?

Estimate two models: compact and augmented

1. Compact model includes only cAge (centered?)
2. Augmented model adds cMiles and cMiles^2 (centered?)
3. Test of coefficient for cMiles^2 indicates if the effect of training miles on races times changes based on number of miles skied.
4. Coefficient for cMiles indicates effect of miles for someone who skies an average number of miles.
5. Model comparison of augmented to compact model provides test of overall Miles effect. NOTE: There is no single coefficient to test. Must use model comparison approach.
6. Test of coefficient for cAge in augmented model provides test of Age effect (controlling for miles).
7. Can re-center Miles on other values to determine linear effect across range of scores. Can quantify effect via model formula using partial derivative.

Example 2: Multiple Regression with Non-Linear Effects

```
mC = lm(Time ~ cAge, data=dMiles)
summary(mC)
Coefficients
             Estimate        SE t-statistic   Pr(>|t|)
(Intercept) 23.55225  0.52421      44.929    < 2e-16 ***
cAge         0.21677  0.04562       4.752 0.00000903 ***
---

Sum of squared errors (SSE): 1714.7, Error df: 78
R-squared:  0.2245
```

```
mA = lm(Time ~ cAge + cMiles + I(cMiles^2),data=dMiles)
summary(mA)
Coefficients
              Estimate         SE t-statistic      Pr(>|t|)
(Intercept) 22.037345  0.481184      45.798      < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000505 ***
cMiles      -0.256454  0.023236     -11.037      < 2e-16 ***
I(cMiles^2)  0.008027  0.001927       4.165 0.0000814131 ***
---

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.7271
```

Example 2: Multiple Regression with Non-Linear Effects

Describe and test the effect of Age on 5K Times, controlling for Miles?

This is obtained from the augmented model. Age has a significant positive effect on 5K race times, controlling for Weekly miles, b= 0.17, ΔR^2= 0.13, t(76)= 6.05, p< .001. For every one year increase in Age, 5K race times increase by 0.17 minutes.

```
modelSummary(mA)
Coefficients: Estimate Std. Error t value Pr(>|t|)
(Intercept) 22.037345   0.481184  45.798  < 2e-16
cAge         0.168096   0.027779   6.051 5.05e-08
cMiles      -0.256454   0.023236 -11.037  < 2e-16
I(cMiles^2)  0.008027   0.001927   4.165 8.14e-05

modelEffectSizes(mA)
Coefficients        SSR df pEta-sqr dR-sqr
(Intercept) 16654.1790  1   0.9650     NA
cAge          290.7416  1   0.3251 0.1315
cMiles        967.2142  1   0.6158 0.4374
I(cMiles^2)   137.7470  1   0.1858 0.0623

Sum of squared errors (SSE): 603.4
Sum of squared total  (SST): 2211.1
```

Example 2: Multiple Regression with Non-Linear Effects

Is there evidence that the effect of Miles is quadratic?
This is tested in the augmented model via the coefficient for $Miles^2$. This coefficient is significant, which indicates that it adds unique variance beyond the linear component. This also means that the size of the linear miles effects changes based on miles (conceptual link to interaction?).

```
Coefficients
              Estimate        SE t-statistic     Pr(>|t|)
(Intercept) 22.037345  0.481184      45.798      < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000505 ***
cMiles      -0.256454  0.023236     -11.037      < 2e-16 ***
I(cMiles^2)  0.008027  0.001927       4.165 0.0000814131 ***
---

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.7271
```

Example 2: Multiple Regression with Non-Linear Effects

Describe and test the overall effect of Miles, controlling for Age?

This is tested via model comparison for the augmented (age, miles, miles^2) vs. compact (age only) models. There is a significant overall effect of miles on 5K times, $F(2,76) = 69.98$, $p < .001$, with weekly mileage accounting for 64.8% of the unexplained variance in 5K times after controlling for age.

```
modelCompare(mC,mA)
SSE (Compact) =  1714.72
SSE (Augmented) =  603.4482
PRE =  0.6480778
F(2,76) = 69.9784, p = 5.820688e-18
```

Example 2: Multiple Regression with Non-Linear Effects

What do you want to report to describe the Miles effect?

1. Size of overall effect in variance terms and sig. test (last slide)
2. b (ΔR^2 or partial eta^2?) and sig tests for linear and quadratic term in centered model
3. Magnitude (and tests) of simples slopes?
4. Overall form of relationship between Miles and 5K times?

Example 2: Multiple Regression with Non-Linear Effects

Describe and test the "average" linear effect of miles.

There was a significant negative linear effect of Miles for skiers with average weekly mileage, b= -0.26, , ΔR^2= 0.44, t(76)= 11.04, p< .001. A one mile increase for runners with average weekly mileage is associated with a 0.26 minute decrease in 5K times.

```
Coefficients
               Estimate        SE t-statistic      Pr(>|t|)
(Intercept) 22.037345  0.481184      45.798      < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000505 ***
cMiles      -0.256454  0.023236     -11.037      < 2e-16 ***
I(cMiles^2)  0.008027  0.001927       4.165 0.0000814131 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.72716
```

Example 2: Multiple Regression with Non-Linear Effects

Report B and sig test for quadratic effect.

However, there was also a significant quadratic effect for Miles, b= 0.01, ΔR^2= 0.06, t(76)= 4.17, p< .001. This indicates that for every one mile increase in weekly mileage, the magnitude of the linear mileage effects decreases.

```
Coefficients
              Estimate         SE t-statistic     Pr(>|t|)
(Intercept) 22.037345  0.481184      45.798      < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000505 ***
cMiles      -0.256454  0.023236     -11.037      < 2e-16 ***
I(cMiles^2)  0.008027  0.001927       4.165 0.0000814131 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.7271
```

Example 2: Simple Effects (Slopes)

```
dMiles$hMiles = dMiles$cMiles - sd(dMiles$cMiles)
mHigh= lm(Time ~ cAge + hMiles + I(hMiles^2),data=dMiles)
Coefficients
               Estimate         SE t-statistic     Pr(>|t|)
(Intercept) 20.026014  0.449986      44.504      < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000505 ***
hMiles      -0.034521  0.058311      -0.592        0.556
I(hMiles^2)  0.008027  0.001927       4.165 0.0000814131 ***
---

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.7271

dMiles$lMiles = dMiles$cMiles + sd(dMiles$cMiles)
mLow= lm(Time ~ cAge + lMiles + I(lMiles^2),data=dMiles)
Coefficients
               Estimate         SE t-statistic           Pr(>|t|)
(Intercept) 27.116838  0.449930      60.269            < 2e-16 ***
cAge         0.168096  0.027779       6.051 0.0000000005048546 ***
lMiles      -0.478387  0.057948      -8.256 0.0000000000000357 ***
I(lMiles^2)  0.008027  0.001927       4.165 0.0000814131068 7 ***
---

Sum of squared errors (SSE): 603.4, Error df: 76
R-squared:  0.7271
```

Example 2: Simple Effects (Slopes)

Report and test simple slopes

As described earlier, for runners with average weekly mileage, a one mile increase is associated with a 0.26 minute decrease in 5K times ($p<.001$).

For runners with low weekly mileage (i.e., 1 SD below mean), a mile increase is associated with a 0.48 minute decrease in 5K times ($p< .001$).

For runners with high weekly mileage (i.e., 1 SD above mean), a mile increase is associated with a non-significant 0.03 minute decrease in 5K times ($p= 0.556$).

Example 2: Simple Effects (Slopes)

```
mrA= lm(Time ~ cAge + Miles + I(Miles^2),data=dMiles)
summary(mrA)
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.862826   1.550650  23.773  < 2e-16
cAge         0.168096   0.027779   6.051 5.05e-08
Miles       -0.736047   0.117271  -6.276 1.96e-08
I(Miles^2)   0.008027   0.001927   4.165 8.14e-05
```

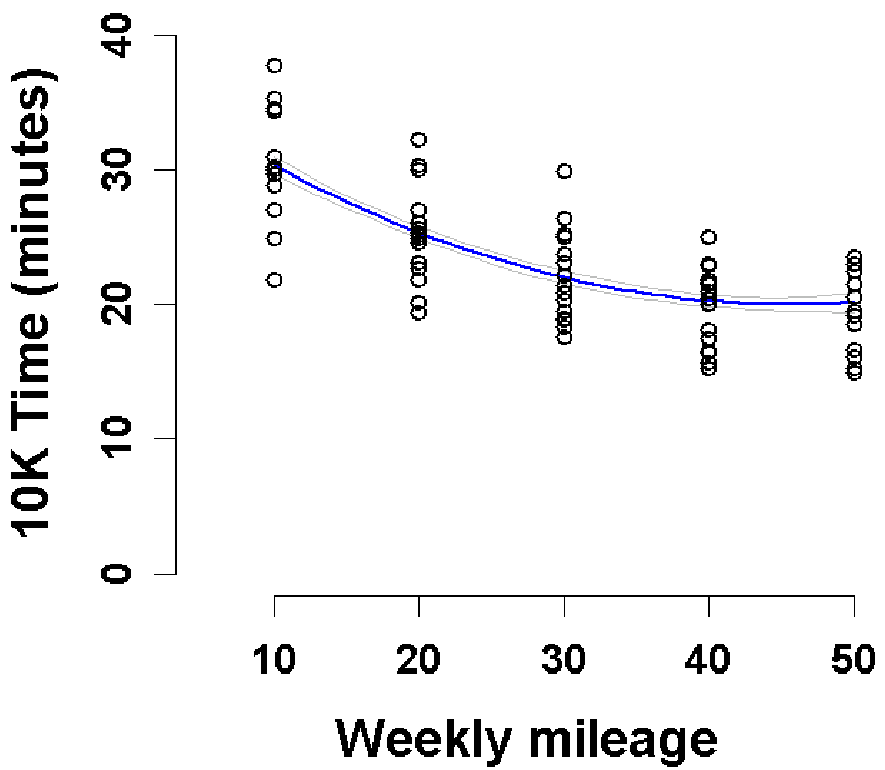
Describe the mileage effect as a formula (Δ5K Time/ ΔMiles)

5K Times= 36.9 + 0.17 * cAge + -0.74 * Miles+ 0.01 * Miles^2

Δ5K Time/ ΔMiles= -.74 + 0.02 * Miles

Example 2: Displaying the Mileage Effect

- Scatterplot with prediction line and error bands for mean Age
- Could graph multiple lines for different ages (what would it look like?)
- Could remove Age variance from scatterplot points (how?)



Multiple Regression with Non-Linear Effects

What would have changed had we: 1) not centered Age or 2) not centered Miles?

Not centering AGE has no effect on AGE or MILES coefficients. There are no higher order (interactive or non-linear) effects involving AGE

Not centering Miles will change Miles coefficient b/c Miles is in a higher order effect involving Miles (i.e., nonlinear $Miles^2$). Miles will be simple effect at 0 on Miles if not centered.

Not centering Miles has not effect on AGE b/c AGE is not in a higher order (interactive) effect with Miles.

Scale of both Age and Miles DOES affect intercept (b_0).


Thank
You
谢谢